

EDUCATION

University of California, San Diego; M.S Electrical & Computer Engineering Fall 2025 - Present

- Concentration:** Intelligent Systems, Robotics & Control
- Related Coursework:** Deep Generative Models, Digital Image Processing, Statistical Learning, Sensing & Estimation in Robotics

Indian Institute of Information Technology, Jabalpur; B.Tech Mechanical Engineering 2019 - 2023

- GPA:** 3.83/4 (Top 5% in batch of 101 students)
- Related Coursework:** Human-centric Robotics, Industrial IoT, Smart Sensors, Design Thinking, Data Structure & Algorithms, C++

TECHNICAL SKILLS

Areas	Robotics, SLAM, Computer Vision, RL, AI, Motion Planning, Force Control Loop, Human-Robot Interaction, Embedded Systems, Sensor Fusion, Deep Learning, VR/AR, Real-Time Systems, Linux, Multi-threading
Programming	C/C++ (14/17), Python, Matlab, Lua, Shell scripting, C#, Data Structures and Algorithms, Design Patterns
Frameworks	ROS/ROS2, TensorFlow, PyTorch, Scikit-learn, Unity, Isaac Lab/Sim, OpenGL, Git/GitHub
Hardware	Quadrupeds, COBOTS, Haptic Devices, Humanoids, Nvidia Jetson Orin, 3D/2D LiDAR, Realsense Camera, STM-32

PROFESSIONAL EXPERIENCE

Addverb Robotics, Robotics Engineer Aug 2023 - Jul 2025

Perception

- Lead Architect for **Object Recognition & 6DOF Pose Estimation** modules, designing and deploying from scratch.
- Designed a **real-time health indicator** for **ORB-SLAM** under test, enabling rapid **SLAM diagnostics** and cutting debug time by 40%.
- Engineered a **self-calibrating camera system** that preserved ± 2 mm accuracy over long durations by correcting for structural droop.
- Built a **Kalman filter-based sensor fusion** module bridging localization gaps between barcodes, easing **brownfield deployment**.

Deep Learning

- Applied **semantic segmentation** to classify SKUs, chargers, and conveyors for improved **robotic perception**.
- Linked **scene understanding with error logs** to generate context-aware diagnostics for rare failures, cutting **debugging time** by 45%.

Systems Optimization

- Dockerized ROS based robotics stack** into modular containers with **CI/CD pipelines** for automated embedded deployment.
- Built an **auto-update workflow** where code pushes generated **Docker images** directly deployed to robots.
- Performed **time-series analysis** for root-cause diagnostics, optimizing **system performance** under real operational constraints.

Deployment

- Led **on-site deployment** for robotic systems with **Siemens, ISITEC, Adidas**, ensuring successful rollout under real-world constraints.
- Implemented **field debugging and rapid iteration**, reducing downtime and accelerating system stabilization post-deployment.
- Integrated **Grafana dashboards** for live monitoring of robotic systems during on-site deployments, improving operational visibility.

RESEARCH EXPERIENCE

Graduate Student Researcher, Advisor: Prof. Michael Yip Sep 2025 – Present

Advanced Robotics and Controls Lab (ARClab), UC San Diego

- Project:** Sim-to-Real Transfer for Dexterous Manipulation of Articulated Tools
- Implementing a **teacher-student framework** targeting **zero-shot sim-to-real transfer** for multi-fingered dexterous manipulation.
- Developed **density-based curriculum learning** with a **safety curriculum** for **collision-aware** grasping in **RL training**.
- Architected the full **NVIDIA Isaac Sim** environment — **MDP**, action/observation spaces, and **reward shaping**.
- Designed structured **reward landscapes** to train stable **RL policies** for multi-fingered dexterous manipulation.

Research Assistant, Neuro-Biohybrid Systems Aug 2025 – Oct 2025

Sanford Consortium for Regenerative Medicine, UC San Diego

- Stood up full **quadruped robot** infrastructure — **teleoperation**, sensor data collection, and **ROS-based** data pipeline.
- Built a **data serialization layer** packaging locomotion telemetry into structured packets for **biological neural network** training.

Research Intern

Jan 2023 – Jul 2023

Defence Research and Development Organisation (DRDO)

- Researched **CNN-based infrared small target detection (IRSTD)** for **early warning missile detection** under low **SNR** conditions.
- Built IR datasets with **pixel-level annotations** and developed **DL pipelines** for noise analysis and **feature aggregation**.

HONORS & ACHIEVEMENTS

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| Published utility patent for the IoT-Enabled 3D Spatial Monitoring Device | 2023 |
| Gold Medalist , IIIT Jabalpur — best cross-disciplinary project among 400+ students | 2023 |
| International Grant Awardee , TIGC La Trobe University — Top 10 of 200+ teams across Asia | 2022 |
| National Winner , L&T Techgium Hackathon — 1st prize of \$12,000, outperforming 30k+ students from 450 universities | 2022 |